

Pre-design solution: Тепличный комплекс «Заря»

Region: Краснодарский край **Greenhouse type:** year-round **Crop:** tomato **Target yield:** 500.0 t/year **Generated:** 2026-06-17 07:36

1. Site inputs and analysis

Target yield per plot area: 25.0 kg/m²/year

Climate parameters (per Russian SP 131.13330):

Parameter	Value
Design winter temperature (t5)	-19.0 °C
Design summer temperature	29.4 °C
Heating degree-days	2510.0
Heating period	149 days
Snow load	1.0 kPa
Wind load	0.38 kPa
Solar radiation (winter)	3.8 MJ/m ² /day

2. Design solution (variant v1)

Rationale: Year-round tomato greenhouse complex in Krasnodar Krai: glass covering was chosen for maximum light transmittance ($\tau \approx 0.88$), which is critical for trellis tomato culture in the southern Russian climate with moderate winter solar radiation. Ridge height is set at 5.5 m to accommodate vertical trellis systems for indeterminate tomato varieties. The complex is arranged as two parallel multi-span block greenhouses, each with six 9 m spans, making efficient use of the 200 × 100 m plot while leaving sufficient inter-block spacing and room for auxiliary infrastructure. Loam soil with groundwater at 3 m depth (above 2 m threshold not triggered) calls for a strip foundation at 1 m embedment with waterproofing precautions. The auxiliary zone exceeds 7 % of the complex footprint and includes all mandatory facilities for a large-scale (>5 000 m²) year-round operation.

Total growing area: 10368 m² **Complex footprint (with auxiliary):** 20000.0 m²

Greenhouse blocks

- **Block A** — 96.0 × 54.0 m (area 5184 m²)
- Heights: eave 3.0 m, ridge 5.5 m

- Layout: block (multi-span), 6 spans of 9.0 m
- Roof: straight, slope 45.0%
- Covering: glass ($\tau=0.88$), glass 4.0 mm - Opaque structures share: 14.0%
- **Block B** — 96.0 × 54.0 m (area 5184 m²)
- Heights: eave 3.0 m, ridge 5.5 m
- Layout: block (multi-span), 6 spans of 9.0 m
- Roof: straight, slope 45.0%
- Covering: glass ($\tau=0.88$), glass 4.0 mm - Opaque structures share: 14.0%

Complex structural parameters

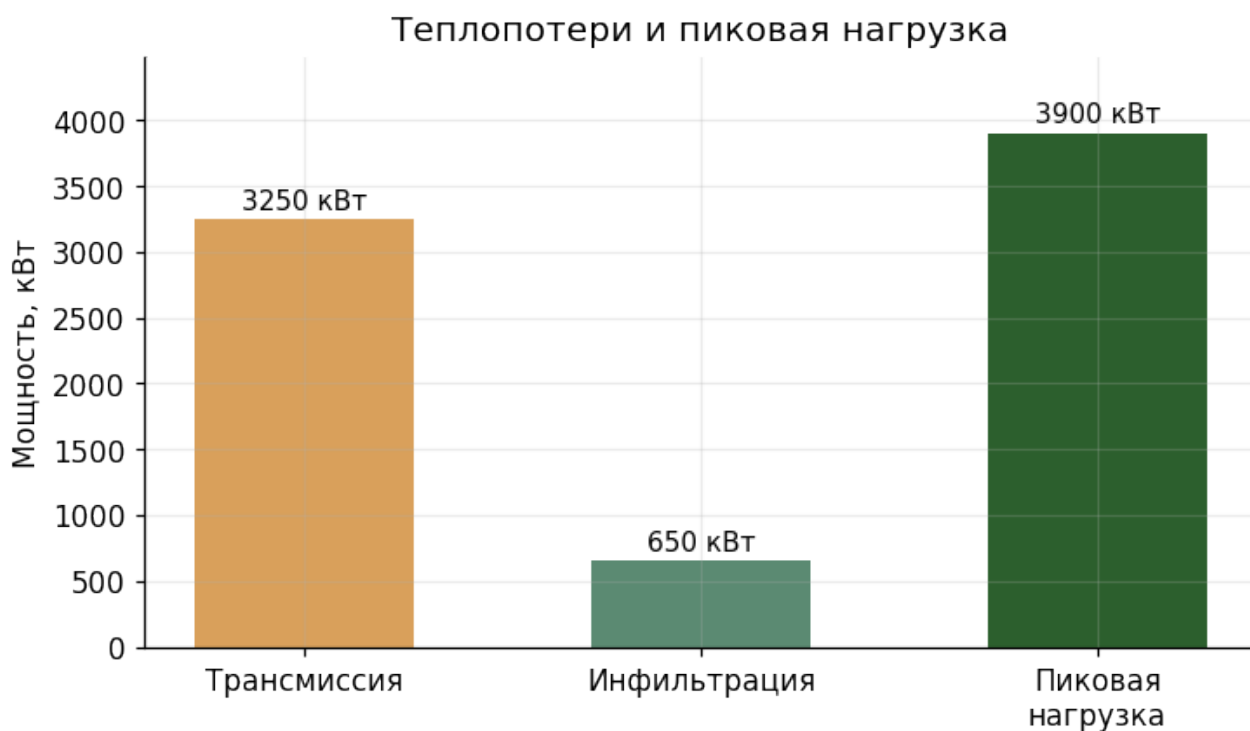
Parameter	Value	SP Clause
Plinth height	0.4 m	5.8
Foundation rise above soil	0.4 m	5.9
Territory fence height	1.8 m	4.16

Auxiliary zones

- **Boiler house** — 120.0 m² (Heating plant and utilities for year-round operation)
- **Tool and fertiliser storage** — 150.0 m² (Storage for tools, agrochemicals and fertilisers)
- **Packaging and sorting** — 200.0 m² (Post-harvest sorting, grading and packaging of tomatoes)
- **Staff facilities** — 80.0 m² (Locker rooms, rest area and sanitary facilities for personnel)
- **Administrative building** — 100.0 m² (Management offices and meeting rooms (complex >5 000 m²))
- **Nutrient mixing and pumping room** — 80.0 m² (Preparation and distribution of nutrient solution for hydroponic systems (complex >5 000 m²))

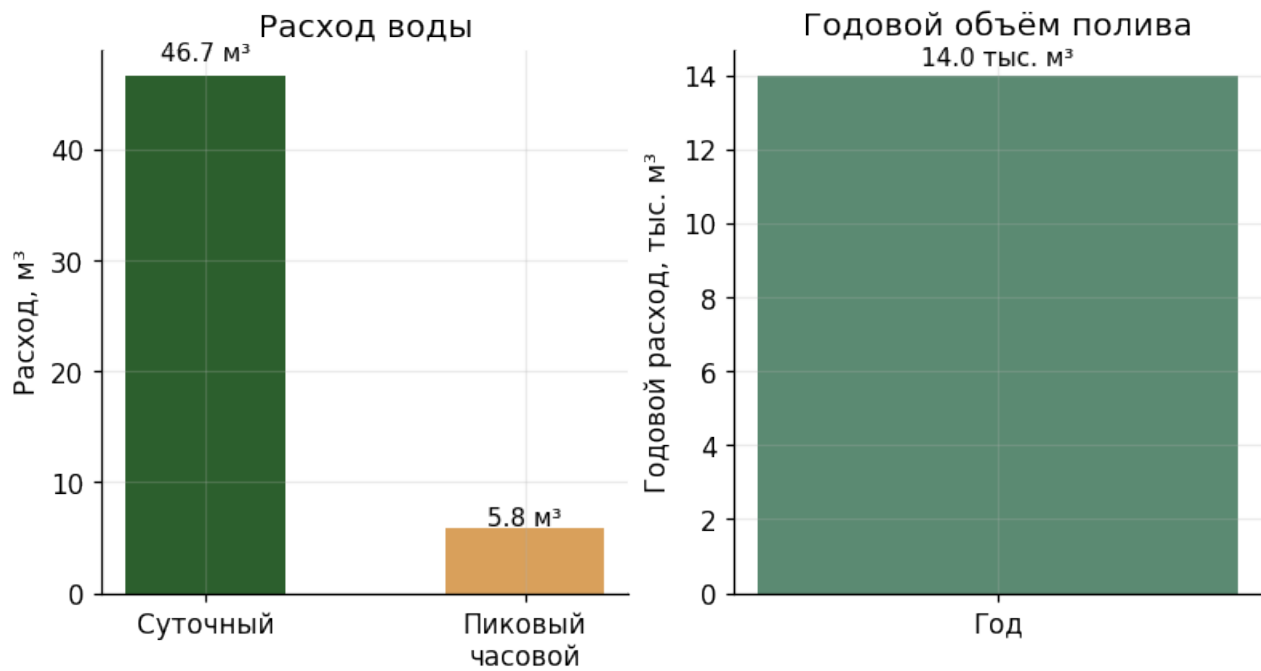
3. Engineering calculations

3.1. Heat supply



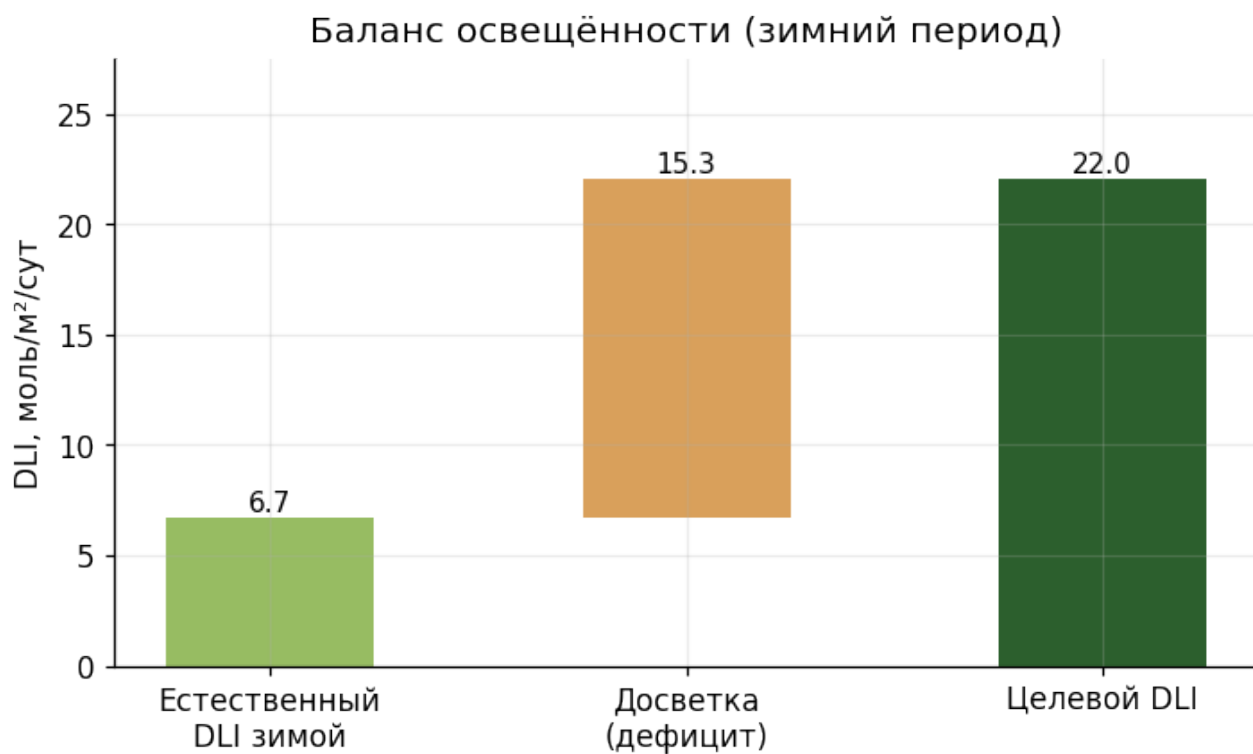
- Design temperature difference: **37.0 °C**
- Envelope area: **13723.2 m²**
- Overall heat transfer coefficient U: **6.4 W/(m²·K)**
- Transmission losses: **3249.7 kW**
- Infiltration losses: **649.9 kW**
- **Peak heating load: 3899.6 kW**
- Annual heat demand: **6348.9 MWh**
- Coolant temperature: **95.0 °C** (SP 7.9 ≤150)
- Lower-zone heat share: **45.0%** (SP 7.13 ≥40)

3.2. Water supply



- Daily demand: **46.66 m³/day**
- Peak hourly: **5.83 m³/h**
- Annual: **13997 m³/year**
- Irrigation method: Drip (default)
- Reliability category: **II** (SP 6.14)
- Hose service radius: **40.0 m** (SP 6.8 ≤45)

3.3. Lighting and supplemental



- Target DLI: **22.0 mol/m²/day**
- Natural winter DLI: **6.7 mol/m²/day**
- **Supplemental lighting required:** installed 210.5 W/m², consumption 5203983 kWh/year
- Aisle floor illuminance: 8.0 lx (SP 8.3 ≤10)

3.4. Annual energy consumption

Годовое энергопотребление: 11553 МВт·ч



3.5. Ventilation

- Summer target air changes per hour: **60.0 h⁻¹**
- Vent opening share of envelope: **20.0%** (SP 7.18 ≥ 20 , ≥ 10 north of 60°N)
- Vent opening share of floor: 26.5%
- Forced ventilation: **not required**

3.6. Structural loads

- Snow load on roof: **16692.0 kN** ($\gamma=1.4$)
- Wind load on walls: **547.0 kN** ($q_{10}=1.0$, $q_2=0.6$)
- Trellis load: 150.0 N/m² ($\gamma=1.3$)

Overload factors and normative values per SP 107.13330 clause 5.14.

3.7. Geotechnical conditions and foundation

- Foundation soil: **loam**
- Groundwater depth: **3.0 m**
- Recommended foundation type: **strip**
- Minimum embedment depth: **1.0 m**

- Perimeter drainage: **not required**
- Waterproofing grade: **standard**

Loam — strip foundation with waterproofing; drainage if groundwater table is high.

Preliminary recommendations. A proper engineering-geological survey per SP 47.13330 is required for working documentation.

4. SP 107.13330 compliance check

Rules checked: **34**

WARNING — ENG.2-aux-share

Подсобные зоны должны занимать 5–30% площади комплекса. Меньше — нехватка для обслуживания; больше — неэффективное использование участка.

- Actual: 3.65
- Required: [5.0, 30.0]
- **Source** (SP 107.13330, original Russian text): Инженерная проверка (не из СП): обоснованная доля подсобок

Generated by agro-greenhouse-designer. Outputs are pre-design estimates and require validation by a qualified engineer before detailed documentation.